

Solving the Exploding Gradient Problem in BPTT

Empirical Analysis of Gradient Norm Clipping in Deep Recurrent Networks

The Exploding Gradient Problem

Training deep RNNs over long sequences (T) exposes vulnerabilities in Backpropagation Through Time (BPTT).

The Mathematical Core:

Jacobian Chain Rule

$$\frac{\partial \mathcal{L}}{\partial h_0} = \frac{\partial \mathcal{L}}{\partial h_T} \prod_{t=1}^T \frac{\partial h_t}{\partial h_{t-1}}$$

- If the largest singular value of the weight matrix $\lambda_{max} > 1$, gradients grow **exponentially**.
- This causes massive L_2 norm deviations, violently destroying optimizer convergence.

To rigorously isolate BPTT instability, we utilized a highly controlled synthetic environment:

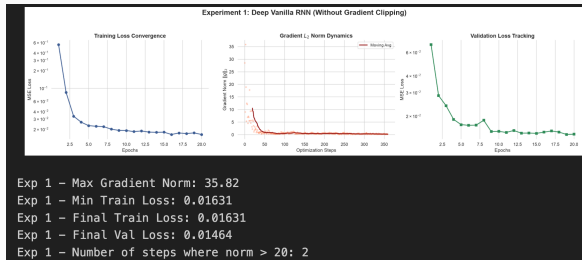
Dataset & Network Configuration

- **Dataset:** Dual-frequency superimposed sine waves with irreducible Gaussian noise.
- **Temporal Depth:** $T = 60$ (Forces backpropagation across 60 dense matrix multiplications).
- **Initialization:** *Strict Orthogonal Weight Initialization* ($\lambda = 1.0$) to guarantee stability at $t = 0$.
- **Optimization:** AdamW with Cosine Annealing Learning Rate.

Exp 1: The Anatomy of an Explosion

Without clipping, eigenvalues naturally drifted > 1 .

- Gradient norms violently spiked to **35.82**.
- Validation loss degraded completely during these massive spikes, undoing epochs of learning.



Exp 2 & 3: Gradient Norm Clipping

Norm Clipping Vector Scaling:

$$g_{new} = g \times \frac{C}{\|g\|_2}$$

Protects step size without distorting gradient direction.

Ablation Study on Threshold (C):

- $C = 0.1$ (**Too Aggressive**): Triggered 360/360 times. Choked the optimizer.
- $C = 5.0$ (**Too Loose**): Triggered only 33 times. Allowed norms up to 75.80.
- $C = 1.0$ (**The Goldilocks Zone**): Cleanly truncated exponential spikes while allowing native gradient flows during stable optimization.

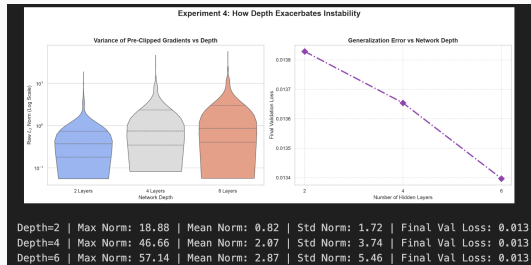
Exp 4: The Depth Penalty

Deeper networks compound Jacobian multiplications across both *spatial* and *temporal* axes.

Geometric Variance Increase

- 2-Layer Std Dev: **1.72**
- 6-Layer Std Dev: **5.46**

Result: $> 3\times$ increase in variance.



Bonus: Architectural Superiority (LSTMs)

Constant Error Carousels (CECs) utilize additive cell states (c_t).

Mathematical Immunity

When the forget gate is saturated, the derivative $\frac{\partial c_t}{\partial c_{t-1}}$ approaches exactly 1.0.

- Both **LSTM** and **GRU** networks massively outperformed Vanilla RNNs.
- They established smooth, deeply converged validation losses natively immune to exponential decay without requiring hyper-aggressive clipping.

For deployment in real-world sequence modeling, we mandate the following architecture stack:

1. **Architecture:** LSTM or GRU to structurally eliminate vanishing gradients.
2. **Safety Net:** Gradient norm clipping with threshold $C \in [0.5, 1.0]$.
3. **Initialization:** Orthogonal initialization for all recurrent matrices.
4. **Optimizer:** AdamW with Cosine Annealing.
5. **Failsafe:** Strict NaN detection during the backward pass for early-stopping.

Questions?

Thank you for your time.